MIL-F-17559C(YD)
12 May 1983
SUPERSEDING
MIL-F-17559B(YD)
4 December 1973

MILITARY SPECIFICATION

# FILTERS, FLUID, PRESSURE, FEEDWATER

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers filters for boiler feedwater which separate oil and suspended solids from raw makeup water and condensate accumulated from steam engines, turbines, and pumps in heating and power steam plants, and which removes organics and chlorine from raw makeup water.

1.2 Classification. Filters covered by this specification shall be of the following types, groups, and styles as specified (see 3.4, 6.2, and 6.8).

Type I - Precoat type.

Group 1 - Vertical tank.

Style A - Vertical leaf or tubular element.
Style B - Horizontal tray.

Group 2 - Horizontal tank.

Style C - Vertical leaf.
Style D - Horizontal plate.

Type II - Chemical type.

Type III - Disk-cartridge type.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4330

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SPECIFICATIONS

FEDERAL

TT-P-664 - Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
VV-L-1071 - Lubricating Oil, Steam Cylinder, Mineral.
PPP-C-2020 - Chemicals, Liquid, Dry, and Paste; Packaging of.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts; Packaging of.
MIL-P-17552 - Pump Units, Centrifugal, Water, Horizontal, General Service and Boiler-Feed: Electric Motor or Steam Turbine Driven.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

Industry Standards and Engineering Information Manual

(Application for copies should be addressed to the American Boiler Manufacturers Association, Suite 700, AM Building, 1500 Wilson Blvd., Arlington, VA. 22209.)



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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
B2.1 - Pipe Threads (Except Dryseal).
B2.2 - Dryseal Pipe Threads.
B15.1 - Safety Standard for Mechanical Power Transmission Apparatus.
B31 - Code for Pressure Piping, Interpretations.
B40.1 - Gauges-Pressure Indicating Dial Type-Elastic Element.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A48 - Gray Iron Castings.
A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
D1253 - Standard Test Methods for Residual Chlorine in Water.
D1888 - Standard Test Methods for Particulate and Dissolved Matter, Solids, or Residue in Water.
D1889 - Standard Test Methods for Turbidity of Water.
D2778 - Standard Test Method for Solvent Extraction of Organic Matter from Water.
D3370 - Standard Practices for Sampling Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII.
Pressure Vessels, Division I.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

HYDRAULIC INSTITUTE (HI)

Standards of Hydraulic Institute

(Application for copies should be addressed to the Hydraulic Institute, 14600 Detroit Ave., Suite 712, North Cleveland, OH 44107.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS-2 - Industrial Control Devices, Controllers and Assemblies.
ICS-6 - Enclosures for Industrial Controls and Systems.
MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)



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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a complete filter assembly for first article inspection and approval (see 4.2.1 and 6.4).

3.2 Standard commercial product. The filters shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product shall be included in the filters being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3.1 Corrosion-resistant steel. Corrosion-resistant steel shall conform to the requirements of ASTM A167.



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3.4 Design and construction. The complete filter system components required shall be as specified (see 6.2). The filter system shall be complete so that when installed and connected to the specified source of electricity (see 3.22) and piping system, it can be used to accomplish the operation for which it was designed. The equipment shall be of such design and construction that the performance requirements specified herein shall be attained when feedwater having the characteristics specified in 3.6.1 and 3.6.3 is processed. Unless otherwise specified (see 6.2), two or more filter tanks shall be required for a single application and shall be arranged for a parallel operation. The equipment shall be designed to withstand strains, jars, vibrations, and other conditions incidental to shipping, storage, and installation. The equipment shall be so designed and arranged that operating, cleaning, and maintenance procedures shall be facilitated. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

3.4.1 Type I. Type I filters shall use a deposited absorbent filter cake for removal of oil and suspended solids from the boiler feedwater, including condensate and raw makeup water. The filter system shall consist principally of (1) a series of metal screens, plates, tubular elements, or filter leaves arranged in parallel in a tank, vertically or horizontally; (2) a support for holding the leaves and for stripping filter cake; and (3) separate tanks for precoat and slurry feed material. Additional components, accessories, pumps, agitators, piping, and fittings required for satisfactory operation of the filter system shall also be furnished. The style of groups 1 and 2 shall be at the manufacturer's option (see 1.2).

3.4.2 Type II. The chemical filter shall be the type that removes oil from the boiler feedwater, including condensate and raw makeup water, by means of chemicals. These chemicals react with oil to form solid particles which, in turn, coagulate into a floc and are deposited on a granular bed through which the feedwater filters. The granular bed also filters out the suspended solids. The filter system shall consist of a filter tank, filter bed, chemical tank, chemical feed pump, means for cleaning the filter bed, and necessary interconnecting piping and fittings of suitable size and characteristics. Additional equipment required for the satisfactory operation of the unit shall be included.

3.4.3 Type III. Type III filters shall be the type using an assembly of disposable disk-cartridges for removal of oil, suspended solids, chlorine or organics as specified (see 6.2), from the boiler feedwater, including condensate and raw makeup water. The filter shall consist of a pressure tank for holding the disk-cartridges and additional components, accessories, piping, and fittings required for the satisfactory operation of the filter system. The feedwater enters the housing and flows over and through each disk-cartridge cell. It is discharged at the inside diameter of the cartridge and then passes down through the center support and out of the filter housing.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts. All threaded parts shall conform to ANSI B1.1, B2.1, and B2.2, as applicable.

3.6 Operating conditions. The filter system shall be used to process feedwater, including condensate and raw makeup water, for a boiler that has a rated



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capacity, steam pressure, and temperature, as specified (see 6.2). The boiler also has an external raw feedwater treatment system, (Zeolite softeners or demineralizers), de-aeration system, and condensate storage tank, and a surge tank for makeup feedwater, as specified (see 6.2). At maximum rating of the boiler, the condensate and surge tank or makeup feedwater system has a minimum net operating volume and time, as specified (see 6.2).

3.6.1 Feedwater temperature and flow. The normal operating temperature of the feedwater will be from 68° Fahrenheit (F) to 180°F, unless otherwise specified (see 6.2). The flow of feedwater through the filter unit in gallons per minute (gpm) shall be as specified (see 3.7 and 6.2).

3.6.2 Operating pressure. Unless otherwise specified (see 6.2), the operating pressure of the filter shall not exceed 90 pounds per square inch gage (psig) for type I and II filters and 125 psig for type III filters.

3.6.3 Feedwater impurities. The feedwater impurities shall not exceed the boiler concentrations as specified (see 6.2).

3.7 Capacity. Filter capacity for type I units shall be based on a flow of feedwater through the filter of not greater than 1 gpm per square foot of filter leaf or element area. The filter capacity for type II units shall be based on a flow of feedwater of not greater than 1/2 gpm per square foot of filter bed area. The filter capacity for type III units shall be based on a flow of feedwater of not greater than 1 gpm per square foot of effective filter media area.

3.8 Performance. When operating at the conditions specified in 3.6 and 3.7, the feedwater effluent quality from the filter system shall not exceed the maximum concentration indicated in table I for oil (organic matter), suspended solids (particulate matter) and free residual chlorine content including filter media, and turbidity. The oil content and suspended solids and free residual chlorine content of the boiler feedwater shall be expressed in milligrams per liter (mg/L). The turbidity shall be expressed in turbidity units as defined in ASTM D1889. Nephelometric turbidity measuring methods shall be used for feedwater with less than 50 turbidity units. When operating with feedwater having less than 10 mg/L of oil and not exceeding the feedwater impurity levels of 3.6.3, the cleaning operation shall be required not more than once every 24 hours for type I and type II units. The filter bed for type II units shall require replacement not more than once every 6 months. The disk-cartridge for type III units shall require replacement not more than once every 3 months. When the feedwater has more than 10 mg/L and a maximum of 30 mg/L of oil content and a suspended solids content no greater than the feedwater impurity levels of 3.6.3, the minimum operating time between cleaning operations and between bed or cartridge replacement shall be as specified (see 6.2). The operating hours shall be based on the filter unit not exceeding a 15 psig pressure differential.

TABLE I. Effluent quality of filtered feedwater.

| Filter | Oil content (mg/L) | Suspended solids (mg/L) | Turbidity (NTU units) | Free residual chlorine (mg/L) |
|---|---|---|---|---|
| All types | 1 max. | 1 max. | 1 max. | |
| Type III | - | - | - | 0.05 |



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3.9 Dimensional requirements. The plan dimensions for location of the filter system shall be as specified (see 6.2). The dimensions of the filter system shall be suitable for the location and allow adequate room for installation, operation, maintenance, and cleaning.

3.10 Type I components.

3.10.1 Type I filter tank. The filter tank for group 1 shall be vertical, and the tank for group 2 shall be horizontal. The filter tank shall be welded carbon steel, equipped with supports for the filter leaves or elements, and arranged to prevent short circuiting or bypassing of the feedwater. Division members between compartments shall be tightly attached to the shell so that no fluid can be bypassed. The tank shall have a feedwater inlet, feedwater discharge, sludge drain, vent, pressure relief device, and gage connections. The filter inlet shall be so baffled that the filter layers shall not be disturbed by inflowing feedwater. The interior arrangement of the tank shall permit cleaning operations and removal of interior components with ease. When specified (see 6.2), the tank shall be equipped with a quick-opening and -closing head. A support designed so that the head may be withdrawn or swung out of the way for leaf removal shall be provided. A support or fixture shall be provided to facilitate handling the leaves when they are removed and returned to the tank. The design and construction of the filter tank shall conform to the ASME Boiler and Pressure Vessel Code for Pressure Vessels, Division 1, Section VIII. The filter tank shall be securely mounted on substantial legs or cradles. When specified (see 6.2), the tank shall be equipped with a filter element so that the used precoat can be separated from the flushing water when the flushing water is discharged to the floor drain.

3.10.2 Filter leaves and elements. The leaf assemblies and elements shall be designed to retain their shape with a differential operating pressure of up to 60 psig. All arrangements specified in styles A through D shall fit into the discharge pipe without leakage to the feedwater effluent. All parts shall be of corrosion-resistant steel construction (see 3.3.1) suitable for the feedwater specified herein.

3.10.2.1 Style A. Vertical filter leaves in group 1 tanks shall consist of heavy-metal screens made up with frames. Vertical, tubular elements shall consist of helically wound wedge-wire screens, circular-woven metal screens held together by longitudinal stiffeners of metal rods, or constructed of circular sintered-metal elements. Vertical filter leaves shall be assembled to a manifold discharge pipe. Vertical tubular elements shall fit into a common horizontal header or tube sheet.

3.10.2.2 Style B. Horizontal filter trays in group 1 tanks shall consist of circular, heavy-metal screens supported by perforated/reinforced metal plates located about a central discharge pipe. The assemblage of leaves and the discharge pipe shall be held in correct relation to each other by a tiebolt or equivalent means independent of the fastening means for any other component.

3.10.2.3 Style C. Vertical filter leaves in group 2 tanks shall consist of heavy-metal screens made up with frames. They are assembled either to a centrally located discharge pipe or to a manifold discharge pipe, normally located at the bottom of the tank.



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3.10.2.4 Style D. Horizontal filter plates in group 2 tanks shall consist of rectangular, heavy-metal screens, supported by perforated/reinforced metal plates and assembled to a manifold discharge pipe.

3.10.3 Precoat tank. A precoat supply tank of welded carbon steel shall be furnished for mixing of the precoat and feedwater. Unless otherwise specified (see 6.2), the volume shall be at least 50 percent greater than the volume of the filter tank. The tank shall be provided with a cover to protect the tank and its contents from contamination. The tank shall be painted or treated to resist corrosion that might occur in operation with the specified feedwater in 3.6.3 and any chemicals as specified (see 6.2), used in treating the feedwater. The tank shall be equipped with a feedwater inlet, precoat mixture discharge, precoat mixture inlet, drain, and liquid level gage connections. Unless otherwise specified (see 6.2), a motor-driven, propeller-type agitator shall be provided, securely attached to, and arranged for ready removal from the tank. When specified (see 6.2), in lieu of a propeller-type agitator, the precoat mixture shall be agitated by circulating it with the feedwater through one or more nozzles. In either case, the agitating means shall provide ample mixing at all times. The propeller and shaft of the agitator shall be constructed of corrosion-resistant steel. Unless otherwise specified (see 6.2), the agitator motor shall be a totally enclosed, nonventilated type, capable of operating on a system of 208 volts, single phase, 60 Hertz (Hz).

3.10.4 Precoat pump. The precoat pump shall be the centrifugal type. All parts in contact with the precoat material and feedwater shall be of corrosion-resistant steel. The pump shall be designed to operate without vapor lock or cavitation and so located that priming will not be necessary. The pump shall be equipped with a pressure-operated bypass. The pump shall be designed to handle at least 10 percent more than the maximum operating pressure at the specified flow of feedwater in 3.6.1. When operating with the conditions specified herein, the discharge head of the pump shall be suitable for the pressure-drop characteristics of the filter system. When specified (see 6.2), the pump shall have sufficient additional capacity to overcome the specified discharge head and suction exterior to the filter system. Pump bearings shall be of adequate proportions to preserve alinement and eliminate excessive looseness and wear, and shall be provided with lubricating fittings of the sealed inlet type. The pump, complete with motor (see 3.22), shall be rigidly mounted to maintain alinement and designed to minimize noise and pulsations in flow. The pump supports shall be ample to withstand the strains imposed by the designed operation and size of piping.

3.10.5 Slurry feed tank. A slurry feed tank of welded steel shall be furnished for the mixing of filter-aid material and water. The capacity of the tank shall be sufficient for 2 days operation under operating conditions specified. The tank shall be painted or treated to resist corrosion by the required filter-aid material and specified feedwater in 3.6.3. A motor-driven, propeller-type agitator shall be provided and arranged so that it can be securely attached to and readily removed from the tank. The propeller and shaft shall be constructed of corrosion-resistant metal. The agitator shall at all times provide ample mixing of the filter-aid material and feedwater. A cover shall be provided to protect the tank and its components from damage and contamination. The tank shall be equipped with a feedwater inlet, slurry discharge, drain, and liquid level gage connections and suitable supports so that it can be secured to foundations. Unless otherwise specified (see 6.2), the agitator motor shall be a totally enclosed, nonventilated type and capable of operating on a system of 208 volts, single phase, 60 Hz.



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3.10.6 Slurry feed pump. The slurry feed pump shall be of the positive displacement type, adjustable for volume discharge, and designed in accordance with the standards of the HI. The pump body shall be of corrosion-resistant steel or of cast iron, meeting the requirements of ASTM A48, class 40. All other parts in contact with the chemicals shall be of corrosion-resistant steel of a type impervious to the chemicals used, filter-aid, water, and air. The pump, complete with motor, shall be mounted on a bedplate adequate to maintain alinement. Pump bearings shall be provided with lubricating fittings of the sealed-inlet type and of adequate proportion to preserve alinement and to eliminate excessive looseness or wear over a normal operating period of 10 years. The pump shall be self-priming and have a built-in bypass valve. Pump capacity shall be at least 10 percent greater than for the maximum operating condition required to meet the specified performance.

3.10.7 Slurry feed pump controls. The slurry feed pump shall be provided with means to permit adjusting the volume pumped. Pump control shall be manual (independent of feedwater flow) or automatic, as specified (see 6.2). Automatic control shall be one of the following methods, as specified (see 6.2):

a. Flow - A preset adjustable relation between the slurry feed volume pumped and the volume of feedwater flowing to the filter and maintain this relationship over a flow variation, as specified (see 6.2).
b. Speed - A preset adjustable relation between the slurry pump speed and the feedwater pump speed and maintain this relationship over a feedwater pump speed range, as specified (see 6.2).
c. On/off - A preset adjustable relation between the slurry feed volume pumped and volume of feedwater flowing. The slurry feed pump shall start and stop corresponding to each start and stop of the feedwater pump. The feedwater pump shall operate at constant speed.

3.10.8 Type I piping and operating accessories. Piping, fittings, strainers, valves, and accessories shall be furnished to facilitate the operations of cleaning, draining, depositing and recirculating the precoat, establishing and maintaining feedwater flow and slurry feed, and all other operations required for proper performance of the equipment. A sight glass shall be provided for viewing the discharge from the filter. Precoat and slurry lines shall be as short and straight as possible and arranged to drain the pumps by gravity. Automatic vents shall be provided for the relief of air from the filter tank. Pressure rated sampling cocks shall be provided for feedwater inlet and discharge. Pump suction connections from tanks shall be designed for a velocity not exceeding 5 feet per second. Pump discharge connections to tanks shall be designed for a velocity not exceeding 10 feet per second. Slurry pump inlet and discharge lines shall be designed for a velocity not less than 1 foot per second. Piping shall be designed to avoid surges in feed that might alternately dislodge filter cake from the leaves or increase the pressure drop by compacting the filter cake. Tank drains shall be located to drain all pockets. The filter tank drain shall be so connected that the feedwater may be drained either by gravity or by means of the precoat pump. Piping, valves, fittings, etc. shall conform to ANSI B31. All pipe threads shall conform to ANSI B2.1. Pressure gages shall conform to ANSI B40.1.

3.10.9 Filter material. Filter material shall be as specified (see 6.2 and 6.6). Sufficient precoat and filter-aid material for two operating cycles shall be furnished with the equipment.



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3.10.10 Flow control. When specified (see 6.2), controls and bypass piping shall be provided to insure a minimum flow of feedwater through the filter at all times. This prevents possible dislodging of the filter cake when feedwater is not supplied to the boiler. When specified (see 6.2), a means shall be provided to balance the flow through each filter so that the alternate filters will maintain staggered cleaning cycles.

3.10.11 Cleaning without disassembly. When specified (see 6.2), filters shall be provided with means for cleaning the leaves or screens without disassembly. Filters so specified shall have the cleaning means arranged for manual operation or automatic control of cleaning cycle, as specified (see 6.2). One of the following means of disengaging the filter cake shall be as specified (see 6.2):

a. Water sprays directed against the cake.
b. A wiper contacting the surface of the plates followed by water working, if required.
c. Backwashing by pneumatic pressure.

3.10.11.1 Manual operation. Filters designed for cleaning by manual operation without disassembly shall be equipped to manually perform the operations of shutting off the feedwater flow to the filter, cleaning the plates, draining or removing the dislodged filter cake from the filter tank, depositing of precoat, re-establishing the flow of slurry, feedwater, and any other necessary operations.

3.10.11.2 Automatic control of cleaning. Automatic control of cleaning shall be initiated as specified (see 6.2). Filters designed for automatic control of the cleaning cycle without disassembly shall be motor-operated. The controls required for automatic operation shall be mounted directly on the control panel of the filter. The filter system shall be equipped to automatically perform the operations of shutting off the feedwater flow to the filter, cleaning of the plates, draining or removing the dislodged filter cake from the filter tank, depositing of precoat, re-establishing the flow of slurry, feedwater, and any other necessary operations. When two or more filters are operating in parallel, a controller shall cause the filters to operate through their cleaning cycle at separate times. When a single filter tank is used, the cleaning cycle shall be completed within the time limit provided by the condensate storage and surge tank capacity, as specified in 3.6.

3.10.12 Filter handling rack. A steel filter rack or support, designed to permit quick and easy transfer of the filter leaves from the filter tank to the handling rack for cleaning, shall be furnished. The rack or support shall be mobile, and so designed that the leaves will be positioned for easy accessibility for hand peeling of filter cake and cleaning of the leaves.

3.10.13 Mobile cart. When specified (see 6.2), a mobile cart, constructed of carbon steel, shall be provided to receive the filter cake peeled from the leaves. The width and length of the cart shall be not less than one and one-half times the diameter of the filter leaf so as to catch the filter cake when peeled by ordinary methods. The depth shall be such that the cake from a complete assembly of leaves can be handled without filling the cart more than 50 percent. The cart shall be equipped with casters and a handle for pulling. The body shall be watertight in construction, and the surfaces shall be painted or treated to resist corrosion that might occur with the feedwater specified herein.



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3.11 Type II components.

3.11.1 Tanks. Filter tanks and chemical tanks shall be of welded steel construction, enclosing, supporting, and protecting all the components and equipped with suitable, removable covers to provide easy access for operation and maintenance. The interior of the tanks shall be accessible for cleaning and removal of components without disengaging connections or fittings. Division members between compartments shall be tight with the shell so that no fluid can be bypassed. Design and construction shall conform to the requirements of the ASME Boiler and Pressure Vessel Code for Pressure Vessels, Division 1, Section VIII.

3.11.1.1 Filter tank. Filter tanks may be vertical or horizontal, and shall provide for feedwater inlet and discharge, filter bed support, sand valve (if required), filter bed wash inlet, filter bed agitation, filter bed cleaning, a means for draining the tank, and filter bed pressure-drop measurement.

3.11.1.2 Chemical tank. The chemical tank shall be provided with a means for easy charging of chemicals and with storage capacity for a minimum 48-hour operation. A mechanical agitator shall be provided to maintain the chemicals in correct condition at all times. The parts in contact with the chemical shall be of corrosion-resistant metal. A level gage shall be provided to indicate the amount of chemicals in the chemical tank.

3.11.2 Filter bed. The filter bed shall be a minimum of 36 inches in depth. It shall consist of no less than 18 inches of granular material approximately 0.016 inch in diameter, supported by granules of increasing size for the remainder of the depth. The design shall prevent the feedwater from bypassing the bed by flow at the sides of the tank. Feedwater shall not channel through the bed under any operative condition. The filter bed shall support the granules and prevent their passage with the feedwater discharge, but shall permit cleaning.

3.11.3 Filter bed cleaning. Mechanical, air, steam, or water agitation and water-wash apparatus shall be provided, as specified (see 6.2). The apparatus shall be complete with motors, valves, nozzles, strainers, and all other necessary components and accessories. Connections for clean water rinse shall be provided to follow the agitation or water wash.

3.11.4 Controls. The filter tank shall be provided with controls for the following sequence of manual operations: (1) feedwater shut-off to permit cleaning of filter bed, (2) bed-cleaning by agitation or backwash, and (3) resumption of feedwater flow with injection of proper proportion of chemicals. Means for periodic scouring of the filter bed, in addition to regular backwashing or agitation, shall also be provided. When specified (see 6.2), automatic operation shall be furnished and shall be initiated by any one of the following methods:

a. Pressure - by a preset adjustable pressure drop across the filter bed.
b. Flow - by a preset adjustable feedwater volume through the filter.
c. Elapsed time - by a preset adjustable operating period.



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3.11.5 Chemical pump. The chemical pump shall be of the positive displacement type. The pump body shall be of corrosion-resistant steel or of cast iron conforming to ASTM A48, class 40. All other parts in contact with the chemicals shall be of corrosion-resistant steel. The pump, complete with motor, shall be mounted on a rigid bedplate adequate to maintain alinement. Pump bearings shall be provided with lubricating fittings of the sealed-inlet type and shall be of adequate proportions to preserve alinement and to reduce wear over a normal operating period of 10 years. The pump shall be self-priming and have a built-in bypass valve. Capacity shall be at least 10 percent greater than that required for the maximum operating condition specified. The chemical pump shall be provided with means to permit adjusting the volume pumped. Pump control shall be manual (independent of feedwater flow) or automatic, as specified (see 6.2). Automatic control shall be by one of the following methods, as specified, (see 6.2):

a. Flow - An adjustable, preset relation between the chemical feed volume pumped and the volume of feedwater flowing to the filter and to main tain this relationship over a flow variation, as specified (see 6.2).
b. Speed - A preset adjustable relation between the chemical pump speed and the feedwater pump speed and to maintain this relationship over a feedwater pump speed range, as specified (see 6.2).
c. On/off - A preset adjustable relation between the chemical feed volume pumped and the volume of feedwater flowing. The chemical feed pump shall start and stop corresponding to each start and stop of the feedwater pump. The feedwater pump shall operate at constant speed.

3.11.6 Piping and operating accessories. Piping, fittings, strainers, valves, and accessories shall be furnished to facilitate the operations of cleaning, draining, replacement of filter media, establishing and maintaining feedwater flow, and all other operations required for proper performance of the equipment. A sight glass shall be provided for viewing the discharge from the filter. Piping, valves, fittings, etc., shall conform to ANSI B31. All pipe-thread connections shall conform to ANSI B2.1. Pressure gages shall conform to ANSI B40.1.

3.12 Type III components.

3.12.1 Filter tank. The filter tank shall be vertical and con- structed with a cast-iron base and cover with a steel, cylindrical shell. The base shall be furnished with an inlet and outlet permitting 90° or parallel piping. Unused connections shall be plugged and used as drains. The base shall be provided with means for mounting the filter. A center post shall provide a means to assemble the disk-cartridge and provide leak-free means for the discharge of the filtered effluent. The filter tanks shall conform to the ASME Boiler and Pressure Vessel Code for Pressure Vessels, Division 1, Section VIII. The tank shall be equipped with a quick-opening and -closing head to permit the easy removal and replacement of the cartridge(s). Gaskets, vents, and pressure gages shall be furnished to facilitate the operation of the filter.

3.12.2 Disk-cartridge. The disposable disk-cartridges shall be an assembly of individual cells. Each cell shall consist of two discs edge-sealed with



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a wet-strength, resin binder around a rigid separator. The center of each cell shall be supported by a spacer to prevent collapse and to assure free flow of the filtered feedwater.

3.12.3 Piping and operating accessories. Piping, fittings, valves, and accessories shall be furnished to facilitate the operations of cleaning, draining, replacement of filters, establishing and maintaining the feedwater flow, and all other operations required for the performance of the equipment. A sight glass shall be provided for viewing the discharge from the filter. Piping, valves, fittings, etc., shall conform to ANSI B31. All pipe-thread connections shall conform to the National taper pipe threads of ANSI B2.1.

3.12.4 Controls. Controls shall permit adjustment of the volume flowing through the filter. When specified (see 6.2), a means shall be provided to balance the flow through each filter so that the alternate filters will maintain staggered cleaning/replacement cycles. Control of the filtration operation shall be manual or automatic, as specified (see 6.2). Control by automatic operation shall be initiated by a preset adjustable pressure drop across the filter elements.

3.13 Feedwater pump. When specified (see 6.2), a centrifugal pump to circulate the feedwater shall be furnished. All pump parts in contact with the feedwater shall be of corrosion-resistant steel. The pump shall be designed to operate without vapor lock or cavitation and so located on the equipment base that priming will not be necessary. The pump shall be equipped with a pressure-operated bypass. The pump shall be designed to operate as near to the maximum efficiency point of the installed impeller as is possible at the flow and head specified in 3.6.1. The rated capacity of the drive motor shall not be exceeded by operation at any point on the pumps curve. The pump shall be designed to handle at least 10 percent more than the specified flow of feedwater in 3.6.1. When operating with the specified conditions herein, the discharge head of the pump shall be suitable for the pressure-drop characteristics of the filter system. When specified (see 6.2), the pump shall have sufficient additional capacity to overcome the specified discharge head and suction exterior to the filter system. Otherwise, the pump shall conform to MIL-P-17552, type I.

3.14 Turbidity indicator. Unless otherwise specified (see 6.2), a photoelectric type turbidity indicator shall be provided to detect the presence of oil and suspended solids in the feedwater discharge from the filter. It will operate an electric alarm circuit when the preset value is exceeded. The instrument shall use nephelometric measuring methods, as defined in ASTM D1889. It shall have at least one scale that measures 0-5 turbidity units. A suitable set of standards and calibration curves shall be furnished. The detector or measuring element shall be capable of mounting in the discharge line of the filter. Unless otherwise specified (see 6.2), the turbidity indicator shall operate on a 120 volt, single phase, 60 Hz system with ground wire. When specified (see 6.2), a battery-operated, portable, turbidity indicator without an alarm shall be furnished.

3.15 Shafts and rods. Operating shafts, rods, and valve stems shall be effectively sealed against leakage and made of corrosion-resistant metal.

3.16 Guards. Guards shall be provided for all moving parts that are dangerous to personnel. Guards shall conform to ANSI B15.1.



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3.17 Seals. Seals for shafts, rods, and valve stems shall be of a type that will not permit sticking to the shaft or rod or react chemically with the shaft or rod. Packings shall be easily adjustable and removable without disassembly of any part except the seal retainer. Seals shall be inert to air, water, oil, and the chemicals handled. Seals shall not require adjustment more than once in 30 days and shall not require renewal more than once a year.

3.18 Control valves. Control valves shall have disks and seats of corrosion-resistant metal. Rubbing and seating surfaces shall be designed to operate without sticking, pitting, or galling. Valve bodies shall be in accordance with ANSI B31.

3.19 Strainers. When specified (see 6.2), strainers shall be provided including where required. The piping shall be so arranged that two or more strainers may be operated in parallel. Pressure drop through any one strainer at rated flow, with specified feedwater, shall not exceed 2 psig.

3.20 Gages. Gages, including a shutoff valve for each gage, shall be supplied to indicate the pressure drop over the filter, suction, and discharge pressure of pumps, and wherever required for proper operation and maintenance of equipment. Unless otherwise specified (see 6.2), gages shall conform to ANSI B40.1, 4.5-inch nominal diameter and shall be mounted on the equipment.

3.21 Control panel. When specified (see 6.2), a NEMA control panel shall be furnished. The panel shall be for wall mounting or for mounting on the equip- ment, as specified (see 6.2). The panel shall contain gages, switches, turbidity indicator, motor controls, alarms, and other controls for the proper operation of the equipment. All control components shall be flush mounted. Each device on the face of the panel shall be identified by a label made of a mate- rial which is durable, corrosion-resistant, and will remain legible for the life of the equipment. All components inside the cabinet shall have permanent type labels. Cabinets shall be of steel construction, welded, and gasketed so as to be dustproof in accordance with NEMA type 12. The panel shall not be subject to being splashed with water during operation or cleaning of the filter.

3.22 Electric motors. Electric motors shall be rated and constructed for continuous duty in accordance with the applicable provisions of NEMA MG 1. Unless otherwise specified (see 6.2), motors shall be totally enclosed, watertight, and ready for operation on a system of 208 volts, 3 phase, 60 Hz. The horsepower rating of each pump motor shall not be exceeded when the pump is delivering the volume of water specified in 3.6, 3.10.4, 3.10.6, 3.11.5, and 3.13, as applicable against the total dynamic head of the intended installation.

3.23 Motor starter. Unless otherwise specified (see 6.2), the across-the-line starter shall include thermal overload protection with a hand/off/automatic switch in its cover and an NEMA type 4 enclosure as defined by NEMA ICS 6. The starter shall be in accordance with NEMA ICS 2 and furnished with capacity and electrical characteristics suitable for the motor according to NFPA No. 70.

3.24 Automatic control. All control and float switches shall bear the UL label or listing mark. Controls shall have the required capacities and electrical characteristics to operate the pumps. All automatic controls and control circuits shall be in accordance with NFPA No. 70.



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3.25 Wiring. Wiring shall be in accordance with NFPA No. 70. Provisions shall be made to ground all electrical equipment.

3.26 Skid bases. When specified (see 6.2), components of the filter system shall be mounted on steel skid bases. The number of bases shall be as determined by the manufacturer. The base construction shall consist of two steel skid runners with an integrally formed, welded, or bolted baseplate for mounting components of the system. The skid runners shall be upturned or beveled at each end to permit skidding the unit into place. The skid runners shall be furnished with mounting holes for anchoring the units of the system to a concrete foundation or wooden deck. The number and size of holes shall be sufficient to anchor the unit under the most severe condition of vibration or thrust likely to develop during operation. The skid runners shall be of sufficient height to provide ready access to any connections located under the units and of sufficient width to provide adequate bearing surface on a smooth, gravel base. Individual mountings shall be furnished, as required, on the baseplates for supporting motors, pumps, and other components in vertical alinement. Other components requiring removal for service, adjustment, or horizontal alinement shall be mounted in a manner to permit such removal or alinement. Tiedown devices and lifting attachments, if furnished, shall be as specified in 3.27.

3.27 Lifting attachments and tiedown devices. When specified (see 6.2), lifting attachments and tiedown devices shall be in accordance with MIL-STD-209.

3.28 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.29 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment (1,000 pounds per square inch or higher) will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.30 Treatment and painting. Unless otherwise specified (see 6.2), the equipment shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the equipment other than corrosion-resisting steel shall be protected against corrosion and shall present a neat appearance.

3.31 Identification marking. Identification shall be permanently and legibly marked directly on the equipment or on a corrosion-resisting metal plate securely attached to the equipment by the manufacturer. Identification plate shall include the manufacturer's model, serial number, manufacturing date, and any other name or trademark to be readily identifiable to the manufacturer.



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3.32 Instruction plates. The filter assembly shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 0.125-inch diameter.

3.33 Workmanship. All equipment being furnished shall be made and finished with a degree of uniformity and grade of workmanship practicable in a well-equipped factory.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).
c. On-site testing (see 4.2.3).

4.2.1 First article inspection. The first article inspection shall be performed on one unit when a first article is required (see 3.1 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4.1, 4.4.2, and 4.4.3. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.1, 4.4.2, and 4.4.3, and the packaging inspection of 4.5. This inspection shall be performed on each unit at the manufacturer's factory facility prior to packaging and shipping. A certified inspection certificate indicating no defects or test failures shall be provided by the manufacturer.

4.2.3 On-site testing. When specified (see 6.2), testing shall be performed on the filter system at the site of the installation. This testing shall be in addition to quality conformance inspection performed at the factory and supplemental to first article inspection, if such inspection is specified. On-site testing shall consist of all tests deemed necessary by the procuring



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activity to verify compliance with the requirements of this specification. On-site testing shall be performed either by the filter manufacturer or by the installing contractor, as specified (see 6.2). The manufacturer shall have the privilege of representation at tests performed by others. When the manufacturer is responsible for on-site tests, the detailed requirements and schedule for the test program will be as specified in the contract (see 6.2). In all cases, deficiencies revealed by on-site testing shall be corrected at the filter manufacturer's expense and any required retesting shall also be at the manufacturer's expense.

4.3 Examination. Each filter unit furnished shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. All units shall be subjected to the following tests as specified in 4.4.1 through 4.4.3.3. Failure to meet these requirements shall constitute cause for rejection.

4.4.1 Hydrostatic tests. All pressure tanks of the filter system shall be hydrostatically tested in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section VIII. The ASME stamp may be accepted as evidence of this test.

4.4.2 Relief valve. The relief valves shall be tested to determine if their relieving capacity and pressure are in accordance with the requirements of the ASME Boiler and Pressure Vessel Code, Section VIII. The ASME stamp may be accepted as evidence of this test.

4.4.3 Performance test. The pressure filter unit shall be operated in accordance with the manufacturer's instructions at maximum capacity using water not exceeding the suspended solids, oil-free residual chlorine content specified in 3.6. The operation for type I or type II filters shall continue until one filter passes through a cleaning cycle and is put back into operation. The type III filters shall have a minimum operating period of 24 hours. When the type I or type II filter goes back into operation or following the minimum operating period for type III filters, an 8 hour test run shall be made. During the test run, oil conforming to Military Symbol 5190 of VV-L-1071 shall be fed by a chemical type feed pump to the throat of a steam ejector furnishing steam to a steam conden- ser so there will be 10 mg/L of oil in the boiler feedwater. Samples of the filtered feedwater discharge shall be withdrawn at the beginning and at approximately 1-hour intervals and consolidated. Sampling and consolidation of the water samples shall be in accordance with ASTM D1888, D2778, and D1253. The composite sample shall then be divided into two parts for testing. Testing shall be in accordance with ASTM D1888, D2778, and D1253.

4.4.3.1 Oil in feedwater discharge. The required water sample for determination of oil in the feedwater discharge of the filter unit shall be tested



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in accordance with ASTM D2778. To be acceptable as evidence of compliance with 3.8, the combined extracts when weighed shall not exceed 1 mg/L of the water sample.

4.4.3.2 Suspended solids and filter media in feedwater discharge. The required water sample for determination of suspended solids in the feedwater discharge of the filter unit shall be tested in accordance with ASTM D1888, Method A. The membrane filter shall be used in filtering the water sample. The test shall be completed only to the determination of the weight of volatile particulate matter and fixed particulate matter on the membrane filter. After treatment of the residue from the membrane filter at 600° centigrade (C) +25°C, the loss in weight of volatile particulate matter shall not exceed 1 mg/L. The weight of the fixed particulate matter on the test crucible shall not exceed 1 mg/L to be acceptable as evidence of compliance with 3.8. The dissolved matter in the filtered solution need not be determined.

4.4.3.3 Free available chlorine in feedwater discharge. The required water sample for determination of chlorine in the feedwater discharge of the filter unit shall be tested in accordance with ASTM D1253, Method A. To be acceptable as evidence of compliance with 3.8, the measured free residual chlorine content of the treated water sample shall not exceed 0.05 mg/L.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

## 5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Disassembly. Items, components, and parts subject to damage or loss, or that would otherwise increase cube of the item at shipment shall be disassembled. Removed nuts, bolts, and washers shall be secured to one of the mating parts. Parts removed and mating parts remaining on the basic unit shall have matchmarking identification indicated on waterproof tags.

5.1.1.4 Unprotected surfaces. Uncoated ferrous metal surfaces, including threaded connections and surfaces exposed by disassembly, shall be coated with type P-1 preservative. Preservatives are not required on painted surfaces and shall not be applied to rubber or rubber-impregnated parts. All threaded connections shall be plugged or capped.

5.1.1.5 Valves and fittings. Valves and fittings, if removed from the equipment, shall be preserved and packaged in accordance with level A requirements of MIL-V-3.



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5.1.1.6 Gages. Gages subject to physical damage shall be wrapped with non-corrosive barrier material and shall have dials and tubes protected with fiberboard or plywood.

5.1.1.7 Piping. Piping, if removed from the equipment, shall be preserved and packaged in accordance with the level A requirements as specified for fittings in MIL-V-3.

5.1.1.8 Electrical equipment. Motor, starter, and electrical controls shall be cleaned, preserved, and packaged in accordance with the applicable level A requirements of MIL-E-17555 and MIL-E-16298.

5.1.1.9 Drive belts. Drive belts shall have tension relieved.

5.1.1.10 Pulleys. Pulleys shall have unpainted surfaces coated with primer conforming to TT-P-664.

5.1.1.11 Exposed chain. Chain shall be immersed in type P-9 preservative, thoroughly drained, and coated with type P-1 preservative. Sprockets shall be coated with type P-1 preservative.

5.1.1.12 Chemicals. Dry chemicals shall be packaged in accordance with the applicable level A requirements of PPP-C-2020.

5.1.1.13 Enclosed gears. Housings shall be filled to the proper level with operating lubricant. Gears shall be operated sufficiently to coat all internal surfaces. Openings into the housing interior shall be sealed with waterproof tape.

5.1.1.14 Pumps. Interior surfaces shall be coated with type P-3 preservative. When practicable, pump interiors shall be coated by flushing while the impeller is rotated.

5.1.1.15 Tools and accessories. Tools and accessories shall be preserved and packaged in accordance with applicable requirements of PPP-P-40, MIL-V-3, MIL-E-17555, MIL-E-16298, and the applicable methods described in MIL-P-116 for parts not specifically covered in the aforementioned specifications.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Packing shall be in accordance with MIL-STD-794. Containers shall be selected from table I for level A requirements.

5.2.2 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.



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## 6. NOTES

6.1 Intended use. The filters are intended for use in power or heating plants that operate up to 800 psig to remove oil and suspended solids (metal oxide and organic matter), chlorine and dissolved organics from the boiler feedwater, including condensate and raw makeup water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and group of unit required (see 1.2).
c. Issue of document required if other than as specified (see 2.1.1).
d. If a first article is required (see 3.1, 4.2.1, and 6.4).
e. The complete filter system components required (see 3.4).
f. When two or more filter tanks are not required and parallel operation not desired (see 3.4).
g. Type of materials type III filter shall remove from feedwater (see 3.4.3).
h. Boiler capacity, steam pressure, and temperature that the filter system will be used for in processing feedwater (see 3.6).
i. Type of feedwater treatment system and its minimum net operating volume and time (see 3.6).
j. If feedwater temperature is to be other than as specified (see 3.6.1).
k. The rate of feedwater flow through the filter unit (see 3.6.1).
l. If operating pressure of the filter is to be other than as specified (see 3.6.2).
m. Boiler concentrations information (see 3.6.3).
n. Minimum operating time between filter cleaning operations and between bed or cartridge replacements when the feedwater has a maximum oil content of 30 mg/L and a suspended solids content not more than allowed by the feedwater impurities of 3.6.3 (see 3.8).
o. Dimensional requirements for location of the filter system (see 3.9).
p. These items apply only to type I components:

(1) If the filter tank is to be equipped with a quick-opening and -closing head (see 3.10.1).
(2) If the filter tank is to be equipped with a filter element for separating the used precoat from the feedwater when draining (see 3.10.1).
(3) What type of feedwater chemical treatment used (see 3.10.3).
(4) If the volume of the precoat tank is to be other than as specified (see 3.10.3).
(5) If the agitator is to be nozzle-type in lieu of propeller-type (see 3.10.3).
(6) When the precoat pump requires additional suction lift and discharge head (see 3.10.4).
(7) If agitator motors are to be other than as specified (see 3.10.3 and 3.10.5).
(8) If the slurry feed pump control is to be manual or automatic (see 3.10.7).
(9) If automatic is specified, the method of control and the flow variation or pump speed range as required (see 3.10.7).
(10) Type of filter material to be used (see 3.10.9 and 6.6).



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(11) If controls and bypass piping are required to insure a minimum flow of feedwater through the filter and if a means is to be provided for balancing the flow through each filter (see 3.10.10).
(12) If a means for cleaning without disassembly is to be provided (see 3.10.11).
(13) If cleaning without disassembly is specified, should control of the cleaning cycle be manual or automatic, and the method of disengaging the filter cake (see 3.10.11).
(14) Acceptable or selections for automatic sequencing initators identified (see 3.10.11.2).
(15) If a mobile cart is to be provided (see 3.10.13).

q. These items apply only to type II components:

(1) Type of filter bed cleaning required (see 3.11.3).
(2) If a means for automatic operation is to be furnished (see 3.11.4).
(3) If the chemical pump control is to be manual or automatic (see 3.11.5).
(4) If automatic is specified, the method of control and the flow variation or pump speed range as required (see 3.11.5).

r. These items apply only to type III components:

(1) If a means is to be provided for balancing the flow through each filter (see 3.12.4).
(2) If control of the filtration operation is to be manual or automatic (see 3.12.4).

s. When feedwater pump is required (see 3.13).
t. When feedwater pump shall have additional capacity to overcome the specified discharge head and suction exterior to the filter system (see 3.13).
u. If the turbidity indicator is to be other than as specified (see 3.14).
v. If the electrical characteristics of the turbidity indicator are to be other than as specified (see 3.14).
w. If a battery-operated, portable, turbidity indicator without an alarm is to be furnished (see 3.14).
x. When strainers are required and where required (see 3.19)
y. If gages are to be other than as specified (see 3.20).
z. If a control panel is to be furnished and whether it is to be mounted on the wall or on the equipment (see 3.21).
aa. If electric motors are to be other than as specified (see 3.22).
bb. If motor starters are to be other than as specified (see 3.23).
cc. If skid base mounting is to be provided (see 3.26).
dd. If lifting attachments and tiedown devices are to be required (see 3.27).
ee. If treatment for fungus resistance is to be required (see 3.28).
ff. If the means for lubrication is to be other than as specified (see 3.29).
gg. If treatment and painting are to be other than as specified (see 3.30).



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hh. If on-site testing is to be required and if it is to be performed by the filter manufacturer or the installing contractor. Detailed requirements and schedule for the test program (see 4.2.3).
ii. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete filter assembly. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Application. Oil in the boiler feedwater from the condensate of steam engines, turbines, pumps, and other power units causes foaming with subsequent carryover of solids. The solids foul the zeolite resins in softeners or demineralizers of the boiler plant system. The use of filters lengthens the operational time of the softeners or demineralizers, increases zeolite resin life, and minimizes the damaging effects of "crud-storms" at startup or from condenser leaks. The filters will reduce the chemical and water requirements because of less frequent regenerations and preserves the ion-exchange capability by removing substances which foul or destroy ion exchange resins. The filters are located upstream of the boiler and any softeners or demineralizers. When oil in the condensate exceeds 10 mg/L (see 3.8), auxiliary equipment may be used to reduce the oil content to 10 mg/L before the feedwater is delivered to the filter. The suspended solids of the feedwater should not exceed the values specified in the ABMA Industry Standards and Engineering Information Manual for the specified operating pressures. If the suspended solids exceed these values, a prefilter of a pressure type using metal screens should be used to reduce them to acceptable levels. The filter may be used for feedwater with oil content to 30 mg/L and greater solids concentration for short periods, but will then require more frequent filter medium renewal and cleaning. Generally, two filter tanks are used in parallel, their size based on calculations in which one-half the maximum boiler feedwater flow is taken through each filter. The entire flow is taken through one filter with dependence on the normal condensate storage to make up the difference during the cleaning cycle of the other. It is possible to use a single filter tank when the condensate system provides sufficient condensate storage capability or is otherwise arranged to allow the filter to be taken off the line for the cleaning cycle and for maintenance. The style B and style D of type I filters are less subject to dislodging of portions of the filter cake when operating with condensate flow that is subject to surges or that is intermittent. Styles A and C of type I filters are more easily cleaned by manual methods. Type III filters are for use with boiler plant systems requiring a minimum of



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relatively clean makeup water where it is desired to reduce the suspended solids with the least equipment and reasonable maintenance and operating costs. The three types of filters are generally available for plants with capacities of 1,000 pounds of steam per hour and greater.

6.6 Filter materials. Manufacturers of filter media can provide assistance in determining what blend of precoat and filter aid best fit the needs of a particular situation. Before making a selection of the filter material, a pilot run or runs should be made with various blends of media and screens to determine the best arrangement to a particular application. Manufacturers make such test filter units available, but may charge for this service.

6.7 Feedwater analysis. The procuring agency should furnish data to the supplier of the filter on the analysis of the feedwater to the filter unit. The analysis of the raw feedwater shall include the following:

| Substance | |
|---|---|
| Calcium | Volatile matter |
| Magnesium | Organic matter |
| Sodium | Total dissolved solids |
| Silica (total) | Suspended solids |
| Silicia - Collodial | Total alkalinity expressed as calcium carbonate |
| Iron oxide and alumina | Carbon dioxide - free |
| Bicarbonate | Hardness expressed as calcium carbonate |
| Carbonate | Free acid |
| Hydroxide | pH |
| Sulphate | Color |
| Chloride | Turbidity |
| Phosphate | |

The data presented should indicate the minimum, maximum, or composite values of the substances found in mg/L. Turbidity should be expressed as NTU units in accordance with ASTM D1889. Sampling and analysis performed shall comply with applicable ASTM procedures. ASTM D3370 should be used as a guide for sampling.

6.8 Cross-reference of classifications.

| MIL-F-17559A(DOCKS) | MIL-F-17559B(YD) and MIL-F-17559C(YD) |
|---|---|
| Type I - Precoat leaf type | Type I - Precoat type |
| Not defined | Group 1 - Vertical tank |
| Class 1 - Vertical leaf | Style A - Vertical leaf or tubular element |
| Class 2 - Horizontal leaf | Style B - Horizontal tray |
| Not defined | Group 2 - Horizontal tank |
| Class 1 - Vertical leaf | Style C - Vertical leaf |
| Class 2 - Horizontal leaf | Style D - Horizontal plate |
| Type II - Chemical type | Type II - Chemical type |
| Not defined | Type III - Disk-cartridge type |



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6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Navy - YD

Preparing activity:

Navy - YD

Project No. 4330-N030



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

4330-N030

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-F-17559C(YD) | FILTERS, FLUID, PRESSURE, FEEDWATER |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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